

# Volatility forecasting - HAR / HARQ / GBM / GBME(GBM+Earning) / GBME(GBM+Earning)+graph (all metrics, DM-bolded)

GBM = gamma-loss Gradient Boosting Machine (HistGradientBoostingRegressor) on own-history features; GBME(GBM+Earning) = GBM + External earnings features; GBME(GBM+Earning)+graph adds the cross-firm correlation-graph neighbour aggregate. QLIKE is the primary loss (equal to the gamma deviance up to a constant); Diebold-Mariano tests use per-trading-date mean loss.

## S&P 500

MSE x1e-7 - RMSE/MAE x1e-4 - QLIKE unscaled. **Bold** on MSE/RMSE/MAE = numerically best (lowest) value in that horizon; **bold** on QLIKE = significantly best by date-clustered Diebold-Mariano (p<0.05).

| h  | Model                   | MSE          | RMSE         | MAE          | QLIKE         |
|----|-------------------------|--------------|--------------|--------------|---------------|
| 1  | HAR                     | 5.754        | 7.586        | 2.132        | 0.3697        |
| 1  | HARQ                    | 5.648        | 7.515        | 2.151        | 0.3780        |
| 1  | GBM                     | 5.557        | 7.454        | 2.089        | 0.3570        |
| 1  | GBME(GBM+Earning)       | <b>5.397</b> | <b>7.346</b> | 2.017        | <b>0.3172</b> |
| 1  | GBME(GBM+Earning)+graph | 5.424        | 7.365        | <b>2.010</b> | 0.3158        |
| 5  | HAR                     | 6.141        | 7.836        | 2.297        | 0.4364        |
| 5  | HARQ                    | 6.132        | 7.831        | 2.309        | 0.4377        |
| 5  | GBM                     | 6.117        | 7.821        | 2.255        | 0.4116        |
| 5  | GBME(GBM+Earning)       | <b>5.972</b> | <b>7.728</b> | <b>2.180</b> | <b>0.3705</b> |
| 5  | GBME(GBM+Earning)+graph | 6.068        | 7.787        | 2.196        | 0.3715        |
| 10 | HAR                     | 6.214        | 7.883        | 2.356        | 0.4615        |
| 10 | HARQ                    | 6.205        | 7.877        | 2.358        | 0.4614        |
| 10 | GBM                     | 6.262        | 7.913        | 2.312        | 0.4307        |
| 10 | GBME(GBM+Earning)       | <b>6.118</b> | <b>7.822</b> | <b>2.239</b> | <b>0.3912</b> |
| 10 | GBME(GBM+Earning)+graph | 6.240        | 7.899        | 2.272        | 0.3939        |
| 22 | HAR                     | 6.327        | 7.954        | 2.419        | 0.4855        |
| 22 | HARQ                    | 6.311        | 7.944        | 2.415        | 0.4831        |
| 22 | GBM                     | 6.233        | 7.895        | 2.359        | 0.4464        |
| 22 | GBME(GBM+Earning)       | 6.098        | 7.809        | <b>2.277</b> | <b>0.4074</b> |
| 22 | GBME(GBM+Earning)+graph | <b>6.087</b> | <b>7.802</b> | 2.300        | 0.4110        |

DM (QLIKE): GBME vs GBM h1 p=0.000 - GBME vs GBM h5 p=0.000 - GBME vs GBM h10 p=0.000 - GBME vs GBM h22 p=0.000 (all significant → GBME QLIKE bolded).

## HOSE

MSE x1e-7 - RMSE/MAE x1e-4 - QLIKE unscaled. **Bold** on MSE/RMSE/MAE = numerically best (lowest) value in that horizon; **bold** on QLIKE = significantly best by date-clustered Diebold-Mariano (p<0.05).

| h | Model                   | MSE          | RMSE         | MAE          | QLIKE         |
|---|-------------------------|--------------|--------------|--------------|---------------|
| 1 | HAR                     | 6.604        | 8.126        | 5.193        | 1.8117        |
| 1 | HARQ                    | 6.389        | 7.993        | 5.062        | 1.7854        |
| 1 | GBM                     | <b>5.399</b> | <b>7.348</b> | <b>3.944</b> | <b>1.5679</b> |
| 1 | GBME(GBM+Earning)       | 5.477        | 7.401        | 3.966        | 1.5730        |
| 1 | GBME(GBM+Earning)+graph | 5.555        | 7.453        | 3.980        | 1.5783        |

| h  | Model                   | MSE          | RMSE         | MAE          | QLIKE         |
|----|-------------------------|--------------|--------------|--------------|---------------|
| 5  | HAR                     | 6.626        | 8.140        | 5.227        | 1.8218        |
| 5  | HARQ                    | 6.519        | 8.074        | 5.152        | 1.8075        |
| 5  | GBM                     | 5.796        | 7.613        | 4.265        | <b>1.6478</b> |
| 5  | GBME(GBM+Earning)       | <b>5.795</b> | <b>7.613</b> | 4.276        | 1.6485        |
| 5  | GBME(GBM+Earning)-graph | 5.815        | 7.625        | <b>4.256</b> | 1.6494        |
| 10 | HAR                     | 6.660        | 8.161        | 5.250        | 1.8277        |
| 10 | HARQ                    | 6.589        | 8.118        | 5.198        | 1.8180        |
| 10 | GBM                     | 5.966        | 7.724        | 4.425        | <b>1.6842</b> |
| 10 | GBME(GBM+Earning)       | 5.962        | 7.721        | 4.441        | 1.6862        |
| 10 | GBME(GBM+Earning)-graph | <b>5.959</b> | <b>7.720</b> | <b>4.407</b> | 1.6837        |
| 22 | HAR                     | 6.716        | 8.195        | 5.282        | 1.8355        |
| 22 | HARQ                    | 6.677        | 8.171        | 5.250        | 1.8299        |
| 22 | GBM                     | 6.211        | 7.881        | 4.615        | <b>1.7263</b> |
| 22 | GBME(GBM+Earning)       | 6.211        | 7.881        | 4.627        | 1.7262        |
| 22 | GBME(GBM+Earning)-graph | <b>6.207</b> | <b>7.878</b> | <b>4.604</b> | 1.7267        |

*DM (QLIKE)*: GBME vs GBM h1 p=0.000 - GBME vs GBM h5 p=0.214 - GBME vs GBM h10 p=0.122 - GBME vs GBM h22 p=0.919 (earnings does not beat GBM → own-history GBM QLIKE bolded).